

BRIAN HUANG

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EDUCATION

University of California, Davis

Davis, CA

Bachelors of Science, Computer Science

September 2021 - June 2025

TECHNICAL SKILLS

Languages: JavaScript/TypeScript, Python, C, C++, SQL, Go, HTML/CSS

Technologies: React, Redux, Next.js, Node.js, Express, Flask, PostgreSQL, Redis, Prisma, AWS, Docker, Kubernetes, CI/CD

Practices: Microservices, REST APIs, Event-Driven Architecture, JWT Auth, OWASP-aligned security, Unit Testing, TDD, Agile, Scrum, SDLC, MVC

RELEVANT EXPERIENCE & PROJECTS

Member in CodeLab @ UC Davis

Sep 2023 - May 2025

React / Node.js / Express / PostgreSQL / WebSockets / Vercel/Render

- Built a **campus-scale Lost & Found web app** in a collaborative engineering team, supporting verified student posts and item discovery.
- Implemented **secure JWT authentication and realtime messaging** via WebSockets, enabling live chat and instant item match alerts.
- Integrated **Google Maps API** to support safe location sharing and structured meetups for item exchanges.
- Led **cloud deployment, testing, and iterative releases** through Agile sprints, standups, and peer code reviews to maintain stability and code quality.

CivicConnect – Smart City Infrastructure Platform

Full-Stack Project

React / Next.js / Node.js / Kafka / PostgreSQL / Redis / Kubernetes / AWS

- Designed and orchestrated **event-driven microservices** processing 1M+ IoT events/day, improving system reliability by 40% and reducing incident response time by 60%.
- Integrated **Redis caching and API load balancing**, accelerating response times by 70% under high throughput.
- Built automated monitoring and alerting pipelines**, improving observability and enabling faster root-cause debugging during peak event volume.
- Deployed services on AWS EKS using CI/CD**, validating performance and fault tolerance before production pushes.

Banking Dashboard - Full Stack Application

Full-Stack Project

TypeScript / React / Node.js / Express / PostgreSQL / Prisma / Redux

- Built **15+ REST API endpoints** using MVC with OWASP-aligned security, input validation, and rate limiting.
- Designed **atomic transactions with Prisma ORM**, preserving correctness across transfers, deposits, and withdrawals.
- Delivered **production-ready features** through Agile sprints, supported by unit testing and structured debugging to ensure operational integrity.

AI Real Estate Valuation Platform

Machine Learning Project

TensorFlow / Scikit-Learn / Optuna / Pandas / CI/CD

- Built a **validated ML training pipeline**, improving model accuracy by 50% using automated preprocessing and hyperparameter optimization.
- Developed **anomaly and outlier detection** to maintain clean training inputs and reduce data noise.
- Deployed with **CI/CD and model interpretability tooling (SHAP/LIME)**, enabling continuous evaluation and drift monitoring post-deployment.